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19 September 1983
SUPERSEDING
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20 August 1969

MILITARY SPECIFICATION

BOILER, STEAM, LOW PRESSURE, FIRETUBE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vertical, steel, firetube, low pressure, light-oil-fired, packaged, boiler-burner units.

1.2 Classification. The boiler shall be of the following styles and sizes, as specified (see 6.2.1):

Style A - With combination float-valve water feeder and low-water cut-off.
Style B - With combination feedwater pump controller and low-water cut-off.

Size 1 - 198,000 British thermal units per hour (Btuh), gross steam output.
Size 2 - 648,000 Btuh, gross steam output.
Size X - Gross steam output as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2.1), the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4520

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SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel.

MILITARY

MIL-V-173 - Varnish, Moisture, and Fungus-Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-P-551 - Pipes and Elbows, Air Conditioning-Heating.
MIL-B-3180 - Boilers and Related Equipment, Packaging of.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-B-18796 - Burner, Single, Oil, Gas, and Gas Oil combination (400,000 Btu's Per Hour and Over Input Capacity).
MIL-B-18797 - Burners, Oil, Mechanical-draft, Automatic (Under 400,000 Btu Per Hour Input Capacity).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461 - Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code.
Section IV - Heating Boilers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C24 - Pyrometric Cone Equivalent (PCE) of Refractory Materials.
D396 - Fuel Oils.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

HYDRONICS INSTITUTE (HI)

Testing and Rating Standard for Heating Boilers.

(Applications for copies should be addressed to the Hydronics Institute, 35 Russo Place, Berkeley Heights, NJ 07922.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Applications for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 296 - Oil Burners.
UL 353 - Limit Controls.
UL 372 - Primary Safety Controls for Gas and Oil-Fired Appliances.
UL 726 - Standard for Oil Fired Burners.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The boilers shall be vertical firetube, low pressure packaged, boiler-burner steam generating units. The boilers shall be furnished complete with oil burners, primary controls, and other accessories and appurtenances specified herein. The boilers shall be shipped with burner equipment, furnace refractory, and controls properly installed on the boiler. Appurtenances, such as the steam gage, safety valve, and gage glass may be shipped disassembled from the boiler to reduce cubage and damage in transit if the manufacturer so elects. All disassembled items must be shipped with the prime equipment.

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3.2 Codes and Standards. The boilers shall conform to the requirements of UL 296 and the ASME Boiler and Pressure Vessel Code, Section IV (hereinafter referred to as the ASME Code), as applicable.

3.2.1 Compliance. Prior to approval of the first shipment, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the boilers he proposes to furnish under this specification meet the requirements of UL 296 and the ASME Code, as applicable.

3.2.2 UL 296 Requirements. Acceptable evidence of meeting the requirements of UL 296 shall be the A certification symbol or label, or a certified test report (see 6.2.2) from a recognized independent testing laboratory indicating the boiler has been tested and conforms to UL 296. Such evidence must be acceptable to the contracting officer.

3.2.3 ASME Requirements. Acceptable evidence of meeting the requirements of the ASME Code shall be the ASME symbol or Form H-2 certification symbol or label, or a certified test report (see 6.2.2) from a recognized independent testing laboratory indicating the boiler has been tested and conforms to the ASME Code. Such evidence must be acceptable to the contracting officer.

3.3 Applicability. Details for equipment construction, design, and performance covered by this specification shall be supplementary to the requirements of the standards cited in 3.2 and compliance with and approval under the UL or ASME standard shall not relieve the supplier of the responsibility for meeting the requirements of this specification.

3.4 First article. When specified (see 6.2.1), the contractor shall furnish a sample boiler for first article inspection and approval (see 4.2.1 and 6.3).

3.5 Standard commercial product. The boiler shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the boiler being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.7 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.8 Design. Boilers shall be designed in accordance with the ASME Code. The boilers shall be designed to generate steam. In accordance with the ASME Code, the design pressure (see 6.4.2) shall be at least 30 pounds-force per square inch, gage (psig) and the maximum operating pressure (see 6.4.3) shall be 15 psig. The boilers shall be designed to withstand the normal strains, jars, and vibrations incidental to shipping, storage, and installation.

3.9 Construction. The bare boiler shall consist essentially of a boiler shell with jacket, a combustion chamber, a tube section, a base, and boiler trim. Boiler construction shall be in accordance with the applicable section of the ASME Code for the pressure specified in 3.8.

3.9.1 Combustion chamber. The boiler shall be furnished with a combustion chamber that is water-cooled at least on the top and sides. The combustion chamber shall be of all-welded, pressure-tight, steel construction. The fire box shall be of either the refractory lined or unlined design. A refractory lined firebox shall be lined with a preformed, castable, or plastic refractory having a Pyrometric Cone Equivalent, as determined by ASTM C24, of not less than 32. Refractory brick will not be acceptable. Unlined furnaces will be acceptable on wet-base boilers in lieu of refractory-lined furnaces provided the unlined furnaces are constructed of stainless steel, type 309, 310, 446, or a type having at least an equivalent scaling resistance. Adequate access to the combustion chamber shall be provided to permit replacement of the refractory and cleaning of the lower tube sheet. An observation port and cover shall be provided to permit visual examination of the flame.

3.9.2 Tubes. Firetubes shall be steel. Tubes shall have an outside diameter of not less than one inch nor more than 2.50 inches. Tube thickness shall be not less than 0.095 inch. Tubes shall be rolled or welded into accurately sized holes in the tubesheets in a manner to provide a water- and gas-tight joint.

3.9.3 Jacket. All exterior surfaces of the boiler, except doors and tappings, shall be covered with not less than one-inch thick batt or blanket-type insulation. Insulation shall have a heat transfer coefficient at a mean temperature of 75° Fahrenheit (°F) of not more than 0.27 Btuh per square foot per inch thickness per degree Fahrenheit temperature difference. The insulation shall be completely enclosed in a sheet metal jacket. The jacket shall have a thickness of not less than 0.029 inch. The jacket and insulation on the top of the boiler shall be designed and installed to allow ready removal for cleaning and replacement of the tubes. The top cover of top discharge boilers shall be capable of supporting the stack specified in 3.13.1.

3.9.4 Base. For refractory lined boilers, the boiler shall be mounted on suitable legs to provide and air space of not less than 2 inches between the bottom of the boiler and the floor. Clearance for wet-base boilers shall be at least 0.5 inch.

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3.9.5 Flue gas outlet. The flue gas discharge outlet shall be located on the top or the rear of the boiler. Unless otherwise specified, (see 6.2.1), the diameter of the flue outlets shall be as follows:

Size 1 - 8-inch.
Size 2 - 12-inch.
Size X - Manufacturer's standard size.

3.9.6 Boiler openings. Boilers shall have not less than three handholes or washout plugs in the lower part of the waterleg and not less than two handholes or washout plugs near the line of the lower tubesheet. Washout plugs shall have a nominal pipe size (NPS) of not less than 1 1/2 inches. The steam connection for size 1 boilers shall not be less than 2-inch NPS; for size 2 boilers, the steam connection shall be not less than 3-inch NPS. The condensate return connection for all size boilers shall be not less than 2-inch NPS. The feedwater inlet shall be baffled as required to prevent direct impingement of the feedwater on surfaces exposed to hot gases.

3.10 Performance. Boilers shall meet the following performance criteria:

a. Maximum average stack temperature - 600° F.
b. Maximum smoke in flue gas - No. 1 on Shell Bacharach scale.
c. Minimum fuel oil firing rate:

(1) Size 1 - 1.9 gallons per hour (gph).
(2) Size 2 - 6.0 gph.
(3) Size X - As specified (see 6.2.1).

d. Maximum heat transfer rate (see 6.4.4) through each square foot of heating surface (see 6.4.5) - 6,610 Btuh.
e. Minimum efficiency - 75 percent overall efficiency (see 6.4.6).

3.10.1 Gross steam output. Gross steam output shall be determined by applicable test methods and procedures of HI Testing and Ratings Standard for Heating Boilers. Minimum gross steam output for size 1 boilers shall be 198,000 Btuh; minimum gross steam output for size 2 boilers shall be 648,000 Btuh; minimum gross steam output for size X boilers shall be as specified (see 1.2 and 6.2.1). Boiler output shall be based on the following conditions:

a. Minimum feedwater temperature - 60°F.
b. Nominal fuel oil heating value - 140,000 Btu per gallon.

3.11 Boiler trim. Each boiler shall be fully equipped with the trim required under the applicable section of the ASME Code plus the additional appurtenances specified herein. The boiler trim, valve, connecting piping and fittings shall be suitable for the design pressure specified in 3.8. The trim appurtenances shall include, but shall not necessarily be limited to the following:

a. An ASME rated safety valve for steam units as required by the ASME Code.
b. A steam pressure gage having a graduated range that will result in the normal pressure indication being within the middle range of the scale.
c. A water-gage glass.

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3.11.1 Feedwater regulator and low water cut-off. The boiler water level shall be controlled and monitored by a feedwater regulator and two low water cut-off switches. The regulator control element shall be of the float-valve type for style A boilers and of the pump-controller type for style B boilers. Float type controls shall be of packless construction with all pivots, levers, and other moving mechanical linkages external to the steam and water space. Each float chamber shall be equipped with a blowdown connection. The feedwater regulator shall meet the requirements of UL 353. The cut-out switches shall be of the float or electrode actuated type. One switch shall be primary having automatic reset on low water and sounding an alarm on high water. The secondary switch shall have a manual reset and shall activate a red indicating light and sound an alarm on low water. The switches may be integral with or separate from the feed water regulator.

3.11.1.1 Style A. For style A boilers, the regulator shall be of the float-actuated type wherein the feedwater flow shall be controlled by the mechanical action of a float-actuated needle valve. The valve needle and seat shall be stainless steel. The float-valve assembly shall be suitable for use with a differential between the water supply pressure and the boiler operating pressure ranging from 10 pound-force per square inch (psi) to 60 psi.

3.11.1.2 Style B. For style B boilers, the regulator shall be of the pump-controller type. The controller shall be of the float-actuated or electrical probe design and shall provide automatic electrical control of the motor starter on a condensate pumping unit in response to a predetermined drop in the boiler water level.

3.11.2 Pressure limit control. The pressure limit control shall be UL-approved, shall be of the manual reset type, and shall be adjustable in the range of at least 2 to 15 psig. The control shall be designed to interrupt both wires of the electrical supply to shut off the burner when the steam pressure reaches the predetermined set point.

3.11.3 Line disconnect switch. A line disconnect switch shall be mounted on each boiler. The switch shall conform to and be installed in accordance with the applicable requirements of NFPA 70.

3.12 Controls.

3.12.1 Size 1. Controls for size 1 boilers shall conform to MIL-B-18797 and shall at least meet with the applicable requirements of UL 372.

3.12.2 Size 2. Controls and control systems shall conform to MIL-B-18796 and shall at least meet the requirements of UL 726. Size 2 boilers shall be furnished with the following control systems:

A. Programming control systems to provide control sequence 1A in accordance with MIL-B-18796. The programming control system shall include repetitive self-checking circuits.

B. Safety control system including flame safeguard control, boiler limit switches, and fuel safety interlocks specified herein and in MIL-B-18796.

C. On-Off combustion control system conforming to MIL-B-18796.

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3.13 Burners. Burners shall be of the high pressure, mechanical, atomizing type conforming to the requirements of MIL-B-18796 or MIL-B-18797, as applicable. The burners shall be designed and adjusted for firing light fuel oils conforming to the following:

a. Fuel oil, grade No. 2 of ASTM D396.
b. Diesel fuel, grades DF-1 and DF-2 of VV-F-800.
c. Turbine fuel, aviation, grade JP-5 of MIL-T-5624.

The burners shall be of the electrically ignited, on-off type having a constant firing rate. Combustion shall be stable and complete at the gross output firing rate.

3.14 Accessories. When specified (see 6.2.1), one or more of the following accessories shall be furnished with each boiler:

a. Smoke pipe.
b. Draft regulator.
c. Lifting attachments.

3.14.1 Smoke pipe. The smoke pipe (stack) shall consist of seven sections of flue pipe, one tee, and an umbrella collar and cap. Each of the seven sections of flue pipe shall be 24 inches long. The stack diameter shall be as specified in 3.9.5 for the flue outlets. The pipe, tee, and umbrella collar and cap shall be zinc-coated steel. The pipe sections shall conform in other respects to MIL-P-551.

3.14.2 Draft regulator. The draft regulator shall be of the automatic, barometric type, designed for installation in the tee furnished with the smoke pipe. The draft regulator shall be adjustable to encompass a range of settings from 0.02 to 0.08 inches of water.

3.14.3 Lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the boiler. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tie down attachments may be identified by stenciling or other suitable marking. Tie down marking shall clearly indicate that the attachments are intended for the tie down of the boiler on the carrier when shipped.

3.15 Electrical requirements. Electrical equipment, enclosures for electrical parts, wiring methods, and oil burner motors shall comply with the requirements of UL 296. The boilers shall be rated for operation on 115-volt, 60-hertz, alternating current and, in accordance with UL 296, shall be capable of operating satisfactorily within the range of 85 to 110 percent of the rated 115 volts. All interconnecting electrical wiring for motors and furnace-mounted controls shall be so furnished and installed that only one connection at the disconnect switch to the electrical supply will be required.

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3.16 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.17 Electromagnetic interference suppression. When specified (see 6.2.1), The unit shall conform to the electromagnetic interference suppression requirements and test limits for class C3, group III equipment as specified in MIL-STD-461.

3.18 Treatment and painting. Unless otherwise specified (see 6.2.1), the boiler shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the boiler other than corrosion-resisting steel shall be protected against corrosion and present a neat appearance.

3.19 Identification marking. Identification shall be permanently and legibly marked directly on the boiler or on a corrosion-resisting metal plate securely attached to the boiler at the source of manufactur. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.20 Technical manuals. When specified (see 6.2.1), the contractor shall provide technical manual(s) conforming to the data item description(s) (DID) listed on the Contract Data Requirement List (CDRL) (DD Form 1423) (see 6.2.2).

3.21 Repair parts. When specified (see 6.2.1), repair parts shall be furnished and shipped with each unit. The repair parts required and the quantity thereof shall be determined in accordance with the provisioning procedures of the contract.

3.22 Workmanship and fabrication. Workmanship and fabrication shall be in accordance with the ASME Code and shall be of such quality as to produce boilers that meet the standards prevalent among manufacturers who normally produce equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standards cited in 3.2. The government reserves the right to examine and test all boilers to determine the validity of the certification.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on the boiler when a first article is required (see 3.4 and 6.2.1). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified (see 4.6 and 6.2.1), the preproduction pack inspection of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.4, the test of 4.5.1, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All boilers of the same style and size offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an acceptable quality level of 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4 Examination. Each boiler selected shall be examined for defects listed in table I.

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TABLE 1. Classification of defects.

Classification	Defects	Requirement Paragraph
Critical:		
01	Safety standards not in accordance with UL 296. (Certification specified in 3.2.2 will be accepted as evidence of compliance.)	3.2
02	Boiler design, and construction not in accordance with the ASME Code. (The Official ASME Code H-Symbol will be accepted as evidence of compliance.)	3.8 and 3.9
03	Safety valve and relieving capacity not in accordance with ASME Code. (Official ASME Manufacturer's Test Report of Safety Valves will be accepted as evidence of compliance.)	3.11
04	Controls not furnished or as specified in 3.12.	3.12
Major:		
101	Boiler not of size required.	1.2
102	Boiler not complete. Component parts missing or not new.	3.9
103	Refractory not furnished for carbon steel furnaces or not of the type specified. (Refractory supplier's specifications will be accepted as evidence of compliance with type specified).	3.9.1
104	Firetubes not of the diameter or wall thickness specified.	3.9.2
105	Insulation not of the type, thickness, or heat transfer coefficient specified.	3.9.3
106	Boiler jacket not of required thickness or construction.	3.9.3
107	Air space under boiler less than specified.	3.9.4
108	Flue gas outlet not of proper size or in correct location.	3.9.5
109	Boiler openings not of the size or number required.	3.9.6
110	Boiler trim not in accordance with ASME Code.	3.9.7
111	Burners and burner equipment not in accordance with referenced standards and specifications.	3.13
112	Accessories not furnished or of the type specified.	3.14
113	Electrical equipment not in accordance with UL296 requirements.	3.15
114	Treatment and painting not as specified.	3.18
Minor:		
201	Marking not as specified.	3.19
202	Specified technical publications missing or incomplete.	3.20
203	Specified repair parts and tools missing or incomplete.	3.21

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4.4.1 Design verification. When specified (see 6.2.1), the manufacturer shall furnish sketches and computations required to verify compliance with the heating surface requirements and heat transfer rates of 3.10(d) when these requirements are applicable.

4.5 Tests.

4.5.1 Hydrostatic test. Each boiler shall be hydrostatically tested in accordance with the ASME Code. Failure of any boiler to pass the test shall be cause for rejection of the boiler.

4.5.2 Performance test. The first article shall be tested to verify compliance with the output and efficiency requirements of 3.10 and 3.10.1 respectively. Tests shall be conducted in accordance with the applicable provisions of the Testing and Rating Standard for Heating Boilers. Test results shall be reported on HI Test Report Sheets. Failure of any boiler to pass the test shall be cause for rejection of the boiler.

4.5.3 Operational test. The boiler shall be tested to verify compliance with the control requirements of 3.12. Conditions of excess steam pressure, flame failure, and low water level shall be initiated during the test run to verify that the applicable safety control acts to initiate a burner shutdown. Failure of the boiler safety control to initiate a burner shutdown shall be cause for rejection of the boiler.

4.5.4 Safety valve test. The pressure limit control shall be locked out or otherwise made inoperative and the boiler safety valves shall be lifted by steam. The relieving capacity, popping pressure, blowdown, and reseating pressure shall be determined by observation and measurement to be in accordance with the ASME Code. This test shall be made in the presence of the Government inspector. The ASME standard symbol will be accepted only as indicating compliance with the design and material requirements of the ASME Code. Failure of the boiler safety valves to relieve the excess pressure in accordance with the ASME Code shall be cause for rejection of the boiler.

4.5.5 Electromagnetic interference suppression tests. To determine conformance to 3.17, the first unit of production shall be subjected to the tests in accordance with MIL-STD-462. When suppressed to conform to 3.17, the manufacturer upon approval of the contracting officer may furnish a certification (see 6.2.2), in lieu of the test, that the boiler meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed. Failure to install the required suppression devices shall be cause for rejection of the boiler.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3180. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2.1), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2.1).

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5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-B-3180 with the level of preservation and the level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Boilers covered by this specification are intended primarily for use at overseas installations to supply low pressure steam for heating purposes. When a condensate pumping unit, as covered by MIL-P-17749, will be connected directly to the boiler for condensate return, style B boilers should be specified and the condensate pumping unit should be equipped with an automatic water make-up valve in lieu of an integrally-mounted motor float switch. Other low pressure boilers are covered by MIL-B-18897.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Style and size required (see 1.2).
c. Gross steam output required, if size X is specified (see 1.2 and 3.10.1).
d. Issue of document required if different than as specified (see 2.1.1).
e. When first article is required for inspection and approval (see 3.4, 4.2.1 and 6.3).
f. When the flue gas discharge outlet shall be other than as specified, (see 3.9.5).
g. Minimum fuel oil firing rate required if size X is specified (see 3.10).
h. When accessories (smoke pipe, draft regulator, or lifting attachments) are required (see 3.14).
i. When fungus and moisture resistance is required (see 3.16).
j. When electromagnetic interference suppression is required (see 3.17).
k. When special treatment or painting is required (see 3.18).
l. When the contractor shall provide technical manuals conforming to data item descriptions (see 3.20 and 6.2.2).
m. When repair parts are required and the parts and quantities to be furnished (see 3.21).
n. When design verification is required (see 4.4.1).
o. When a preproduction pack inspection is required and the time frame required for submission (see 4.6).
p. Level of preservation and level of packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations are invoked and the DD Form 1423 is not used, the data specified below shall be

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delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.
3.2.2, 3.2.3, and 4.5.5	Certification Data/Report	UDI-A-23264B
3.19	Publications, Commercial	DI-M-24006B
4.4.1	Engineering Drawing	DI-E-7031

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete boiler. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Definitions. The following definitions shall be mandatory for use by the supplier in interpreting and implementing the requirements of this specification.

6.4.1 Packaged boiler. A packaged boiler is a combination of a boiler, burner, combustion chamber design, and controls all factory-assembled and prewired to form complete units ready for operation when connected to a stack and to fuel, water, and electrical service.

6.4.2 Design pressure. The pressure used in the design of a boiler for the purpose of determining the minimum permissible thickness or physical characteristics of different parts of the boiler.

6.4.3 Operating pressure. The operating pressure of a boiler is the pressure at which the boiler normally operates. The operating pressure is usually kept at a suitable level below the setting of the pressure-relieving devices to prevent frequent opening of the devices.

6.4.4 Heat transfer rate. The heat transfer rate shall be determined by dividing the gross steam output by the heating surface.

6.4.5 Heating surface. Heating surfaces include surfaces of the boiler exposed to the products of combustion on one side and water on the other, expressed in square feet. For the purposes of this specification, the heating surface shall be measured on the side receiving the heat and shall include only the surface up to the mid-point of the gage glass.

6.4.6 Overall efficiency. Overall efficiency is the measured heat output of the boiler divided by the measured heat input to the boiler under steady state conditions.

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6.5 Supersession data. The classification of boilers in this document differs from the superseded specification in the following respects:

MIL-B-16220D	MIL-B-16220E
Style A - With combination float-valve water feeder and low water cut-off.	Style A - No change
Style B - With combustion feedwater pump controller and low water cut-off.	Style B - No change
Size 1 - 198,000 Btu/h	Size 1 - 198,000 Btuh, gross steam output
Size 2 - deleted	Size 2 - 648,000 Btuh, gross steam output
Size 3 - 648,000 Btu/h	Size 3 - deleted
Size X - Gross steam output, as specified	Size X - No change

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

Navy - YD

Project No. 4520-0281

Review activity:

Air Force - 84

User activity:

Army - CE
Navy - MC

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side) Project No. 4520-0281

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-16220E	BOILER, STEAM, LOW PRESSURE, FIRETUBE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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